ASSIGNMENT 4

Q10: Student Result Manager (Additional Practice)

Instructions: Save JavaScript-only solutions as `.js` files and run with Node.js. For form questions, save the full code as `.html` and open it in a browser. No frameworks are used.

Solution

```
// Additional practice question to provide 10 separate PDFs.
class Student {
  constructor(name, marks) { this.name = name; this.marks = marks; }
  getTotal() { return this.marks.reduce((sum, mark) => sum + mark, 0); }
  getAverage() { return this.getTotal() / this.marks.length; }
  getGrade() {
    const avg = this.getAverage();
    if (avg >= 90) return "A";
    if (avg >= 75) return "B";
    if (avg >= 60) return "C";
    if (avg >= 40) return "D";
    return "F";
  }
}
const students = [
  new Student("Aman", [88, 92, 81]),
  new Student("Priya", [76, 84, 79]),
  new Student("Karan", [45, 52, 48]),
  new Student("Meera", [32, 38, 29])
];
students.forEach(s => console.log(
  `${s.name}: Total=${s.getTotal()}, Average=${s.getAverage().toFixed(2)}, Grade=${s.getGrade()}`
));
console.log("Passed:", students.filter(s => s.getAverage() >= 40).map(s => s.name));
```

Expected result / testing

Run with Node.js to see each student’s total, average and grade, plus a filtered list of students with average marks of at least 40.